Problem A. Next Even Integer

Time Limit 1000 ms
Mem Limit 131072 kB

Integer n is given. Print the next even integer for n.

Input

One integer n.

Output

Print the next even integer after n.

Examples

Input	Output
7	8

Input	Output
4	6

Problem B. Watermelon

Time Limit	1000 ms
Mem Limit	65536 kB
Input File	stdin
Output File	stdout

One hot summer day Pete and his friend Billy decided to buy a watermelon. They chose the biggest and the ripest one, in their opinion. After that the watermelon was weighed, and the scales showed w kilos. They rushed home, dying of thirst, and decided to divide the berry, however they faced a hard problem.

Pete and Billy are great fans of even numbers, that's why they want to divide the watermelon in such a way that each of the two parts weighs even number of kilos, at the same time it is not obligatory that the parts are equal. The boys are extremely tired and want to start their meal as soon as possible, that's why you should help them and find out, if they can divide the watermelon in the way they want. For sure, each of them should get a part of positive weight.

Input

The first (and the only) input line contains integer number w ($1 \leq w \leq 100$) — the weight of the watermelon bought by the boys.

Output

Print YES, if the boys can divide the watermelon into two parts, each of them weighing even number of kilos; and NO in the opposite case.

Examples

Input	Output
8	YES

Note

For example, the boys can divide the watermelon into two parts of 2 and 6 kilos respectively (another variant — two parts of 4 and 4 kilos).

Problem C. New Year Resolution

Time Limit	1000 ms
Code Length Limit	50000 B
OS	Linux

To start the year 2026 off, Chef made a resolution to exercise daily. He decided to do exactly X push-ups every day.

If he sticks to his resolution, how many push-ups will he do in the month of January?

Note that the month of January has 31 days.

Input Format

- The first and only line of input will contain a single integer X, denoting the number of push-ups Chef does every day.

Output Format

Output a single integer: the number of push-ups Chef will do in January.

Constraints

- $1 \leq X \leq 100$

Sample 1

Input	Output
5	155

Doing 5 push-ups every day, Chef will end up doing $5 \times 31 = 155$ push-ups in January.

Sample 2

Input	Output
100	3100

Problem D. Sum

Time Limit	2000 ms
Mem Limit	65536 kB
OS	Windows

Your task is to find the sum of all integer numbers lying between 1 and N inclusive.

Input

The input consists of a single integer N that is not greater than 10000 by it's absolute value.

Output

Write a single integer number that is the sum of all integer numbers lying between 1 and N inclusive.

Sample

Input	Output
-3	-5

Problem E. Team

Time Limit	2000 ms
Mem Limit	262144 kB
Input File	stdin
Output File	stdout

One day three best friends Petya, Vasya and Tonya decided to form a team and take part in programming contests. Participants are usually offered several problems during programming contests. Long before the start the friends decided that they will implement a problem if at least two of them are sure about the solution. Otherwise, the friends won't write the problem's solution.

This contest offers n problems to the participants. For each problem we know, which friend is sure about the solution. Help the friends find the number of problems for which they will write a solution.

Input

The first input line contains a single integer n ($1 \le n \le 1000$) — the number of problems in the contest. Then n lines contain three integers each, each integer is either 0 or 1. If the first number in the line equals 1, then Petya is sure about the problem's solution, otherwise he isn't sure. The second number shows Vasya's view on the solution, the third number shows Tonya's view. The numbers on the lines are separated by spaces.

Output

Print a single integer — the number of problems the friends will implement on the contest.

Examples

Input	Output
3 1 1 0 1 1 1 1 0 0	2

Input	Output
2 1 0 0 0 1 1	1

Note

In the first sample Petya and Vasya are sure that they know how to solve the first problem and all three of them know how to solve the second problem. That means that they will write solutions for these problems. Only Petya is sure about the solution for the third problem, but that isn't enough, so the friends won't take it.

In the second sample the friends will only implement the second problem, as Vasya and Tonya are sure about the solution.

Problem F. Theatre Square

Time Limit	1000 ms
Mem Limit	262144 kB
Input File	stdin
Output File	stdout

Theatre Square in the capital city of Berland has a rectangular shape with the size $n \times m$ meters. On the occasion of the city's anniversary, a decision was taken to pave the Square with square granite flagstones. Each flagstone is of the size $a \times a$.

What is the least number of flagstones needed to pave the Square? It's allowed to cover the surface larger than the Theatre Square, but the Square has to be covered. It's not allowed to break the flagstones. The sides of flagstones should be parallel to the sides of the Square.

Input

The input contains three positive integer numbers in the first line: n, m and a ($1 \le n, m, a \le 10^9$).

Output

Write the needed number of flagstones.

Examples

Input	Output
6 6 4	4

Problem G. Elephant

Time Limit 1000 ms
Mem Limit 262144 kB

An elephant decided to visit his friend. It turned out that the elephant's house is located at point 0 and his friend's house is located at point $x (x > 0)$ of the coordinate line. In one step the elephant can move $1, 2, 3, 4$ or 5 positions forward. Determine, what is the minimum number of steps he need to make in order to get to his friend's house.

Input

The first line of the input contains an integer x $(1 \le x \le 1\,000\,000)$ — The coordinate of the friend's house.

Output

Print the minimum number of steps that elephant needs to make to get from point 0 to point x.

Examples

Input	Output
5	1

Input	Output
12	3

Note

In the first sample the elephant needs to make one step of length 5 to reach the point x.

In the second sample the elephant can get to point x if he moves by $3, 5$ and 4. There are other ways to get the optimal answer but the elephant cannot reach x in less than three moves.

Problem H. Divide the Apples

Time Limit 1000 ms
Mem Limit 131072 kB

n schoolchildren divide k apples evenly, the residue remains in the basket. How many apples remain in the basket?

Input

Two positive integers n and k not greater than 1500 — rarely are there more pupils in school, and where to find such a basket?

Output

Print the number of apples in the basket.

Examples

Input	Output
3 14	2

Input	Output
10 100	0

Problem I. Squid Game

Time Limit	1000 ms
Code Length Limit	50000 B
OS	Linux

In the deadly "Squid Game," the participants start with N players. After the game ends, only K players survive. The prize pool increases based on the number of players eliminated.

Each eliminated player contributes a fixed amount of $10,000$ units to the prize pool.

Your task is to calculate the total prize money added to the pool, given the number of players N at the start of the game and the number of players who survive, K.

Input Format

The first line contains two integers N and K, where:

- N is the total number of players at the start of the game.
- K is the number of players still alive after the game.

Output Format

Print a single integer — the total prize money added to the pool.

Constraints

- $1 \leq K < N \leq 100$

Sample 1

Input	Output
`10 5`	`50000`

The game starts with 10 players, and 5 of them remain in the end. This means 5 players were eliminated, so the total prize pool is $5 \cdot 10000 = 50000$.

Sample 2

Input	Output
`89 33`	`560000`

The game starts with 89 players, and 33 of them remain in the end. This means 56 players were eliminated, so the total prize pool is $56 \cdot 10000 = 560000$.

Problem J. The 67th Integer Problem

Time Limit 1000 ms
Mem Limit 262144 kB

Welcome to the New World, O Esteemed Earthly Visitor. You've been summoned by Macaque, a primate with four legs, a god complex and a terminal addiction to the word "trivial". You are Undertaking a Journey of Great Importance. Such Incredible Importance. No Journey Will EVER be as IMPORTANT as this one (and nothing this narrator says will sound so distinctly... orange again). You are implored to cooperate with Macaque, for his wrath (and joblessness) are unending. There is no room for error or incompetence. Lousiness shall be met with the full force of the law.

Macaque is given an integer x. Your task is to choose an integer y such that the value of $\min(x, y)^*$ is **maximized**.

If there are multiple valid y, you may output any of them.

$^*\min(x, y)$ is defined as the minimum of integers x and y.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 6767$). The description of the test cases follows.

The only line of each test case contains a single integer x ($-67 \le x \le 67$).

Output

For each test case, output one integer y ($-67 \le y \le 67$) such that $\min(x, y)$ is maximized.

Examples

Input	Output
3 1 3 5	2 4 6

Note

In the first case, 2 is a possible answer because $\min(1, 2) = 1$, which can be shown to be maximal.